

PROJECT REPORT

Numerical Investigation of an Angle Crack Embedded in a Plate Using COMSOL Multiphysics

A Fracture Mechanics Study Based on the NAFEMS Angle-Crack Benchmark

Prepared from the supplied COMSOL Multiphysics model

Model file: angle_crack_plate.mph

COMSOL Multiphysics 6.3 (Build 254)

Name: Akhilesh Yadav

Roll No.: 251050004

Department: Mechanical Engineering

Institution: Indian Institute of Technology Kanpur

Abstract

This project presents a two-dimensional finite-element investigation of a rectangular plate containing a centrally embedded crack inclined at different angles to the tensile loading direction. The analysis is based directly on the supplied COMSOL Multiphysics model, `angle_crack_plate.mph`. The model uses the Solid Mechanics interface with linear isotropic elasticity and evaluates the energy release rate through the J-integral at both crack tips. Mode-I and Mode-II stress intensity factors are obtained in normalized form relative to $K_0 = \sigma\sqrt{\pi a}$. A parametric study is performed for crack angles $\beta = 90^\circ$, 67.5° , and 22.5° . The calculated results show that the horizontal crack ($\beta = 90^\circ$) is predominantly Mode I, whereas reducing the crack angle introduces a progressively stronger Mode-II contribution. The computed normalized stress intensity factors agree closely with the NAFEMS reference values, with the largest average absolute deviation among the reported nonzero Mode-II cases being about 1.86%. The results demonstrate the suitability of the J-integral and stress-intensity-factor approach for numerical verification of mixed-mode fracture mechanics problems.

Keywords: fracture mechanics, angle crack, J-integral, stress intensity factor, Mode I, Mode II, COMSOL, NAFEMS.

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1. Introduction

Cracks and other defects are among the most important causes of structural failure in engineering components. In linear elastic fracture mechanics (LEFM), the local field near a crack tip is commonly characterized using the stress intensity factors K_I and K_{II} , representing opening and sliding fracture modes, respectively. An alternative but closely related measure is the J-integral, which represents the energy release rate associated with crack extension.

An inclined crack subjected to remote tension produces mixed-mode fracture behavior. The proportion of Mode I and Mode II loading depends strongly on crack orientation. This makes an angle-crack plate a useful verification problem for finite-element fracture mechanics implementations. The NAFEMS R0020 benchmark series includes an angle crack embedded in a plate specifically for assessing numerical correctness of linear elastic fracture mechanics calculations.

The supplied COMSOL model reproduces this benchmark using a 2D rectangular plate, a centrally located crack, a tensile boundary load, elastic material behavior, a parametric sweep of crack angle, and J-integral evaluation at both crack tips.

2. Objectives

- Develop and document the finite-element model of a centrally embedded angle crack in a rectangular plate.
- Study the effect of crack angle on the J-integral and normalized stress intensity factors.
- Determine the relative contributions of Mode I and Mode II fracture loading.
- Compare the numerical stress intensity factors with the NAFEMS benchmark reference values.
- Assess the consistency of the results at the two crack tips and identify the effect of nonsymmetric boundary loading.

3. Theoretical Background

3.1 Stress Intensity Factors

For a crack subjected to linear elastic loading, the near-tip stress field can be represented in terms of stress intensity factors. K_I characterizes the crack-opening Mode-I field, while K_{II} characterizes the in-plane sliding Mode-II field. For the present benchmark, the model normalizes these quantities using

$$K_0 = \sigma \sqrt{(\pi a)}$$

where σ is the applied nominal tensile stress and a is the half crack length. In the supplied model, $\sigma = 100$ MPa and $a = 25$ mm, giving $K_0 = 2.8025 \times 10^7 \text{ Pa}\sqrt{\text{m}} = 28.025 \text{ MPa}\sqrt{\text{m}}$.

3.2 J-Integral

The J-integral is an energy-based fracture parameter. For a homogeneous linear elastic material, it is related to the crack-driving force and can be evaluated using a contour surrounding the crack tip. A key advantage of the J-integral is that, when the numerical solution is sufficiently accurate, its value is relatively insensitive to the exact contour selected around the crack tip.

3.3 Mixed-Mode Fracture

When the crack is perpendicular to the tensile direction, the loading is essentially opening dominated and KII approaches zero. As the crack is inclined, the local traction components acting on the crack faces produce a sliding component. Thus, decreasing β from 90° to 22.5° changes the fracture mechanism from Mode-I dominated toward Mode-II dominated behavior.

4. Problem Definition

The model consists of a rectangular 2D plate with a centrally embedded crack. The rectangle is defined by width $2b$ and height $2h_0$, while the crack length is $2a$. The crack is initially created as a horizontal line from $(-a, 0)$ to $(a, 0)$ and then rotated according to the parameter β . Three crack angles are analyzed.

Parameter	Expression / Value	Description
b	50 mm	Half width of plate
h_0	$1.25\ b = 62.5\ \text{mm}$	Half height of plate
$2b$	100 mm	Total plate width
$2h_0$	125 mm	Total plate height
a	$0.5\ b = 25\ \text{mm}$	Half crack length
$2a$	50 mm	Total crack length
β	$90^\circ, 67.5^\circ, 22.5^\circ$	Crack angle
σ	100 MPa	Applied tensile load
E	207 GPa	Young's modulus
ν	0.30	Poisson's ratio
K_0	$28.025\ \text{MPa}\sqrt{\text{m}}$	Reference normalization factor

4.1 Model Geometry

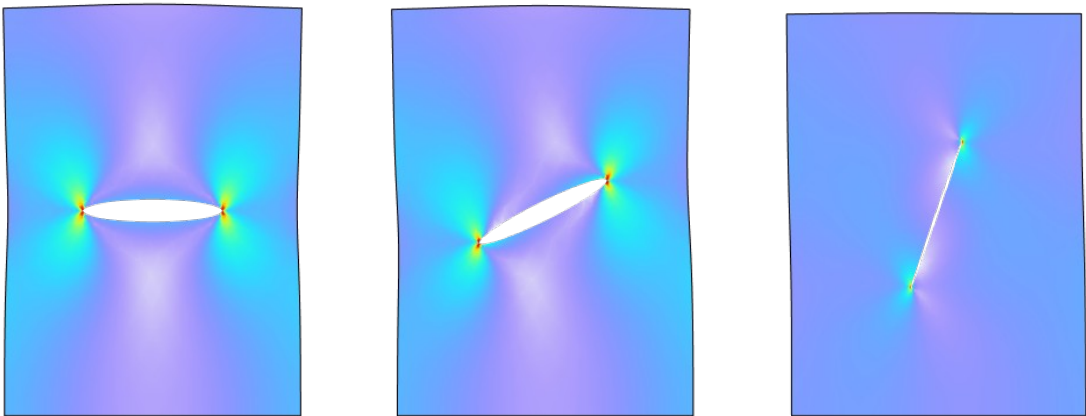


Figure 1. Stress-field visualizations stored with the supplied COMSOL model for $\beta = 90^\circ$, 67.5° , and 22.5° .

The stored model image clearly shows the stress concentration localized at the crack tips. The change in crack orientation also changes the distribution of the surrounding stress field and the relative severity of the two fracture modes.

5. Numerical Model in COMSOL Multiphysics

5.1 Physics Interface and Material

The supplied file uses the Solid Mechanics interface in a two-dimensional component. The constitutive model is linear elastic, isotropic, and defined by Young's modulus $E = 207 \text{ GPa}$ and Poisson's ratio $\nu = 0.3$. The density stored in the material definition is 8000 kg/m^3 ; density does not influence the reported stationary elastic response in this static analysis.

5.2 Crack Representation

The crack is introduced as a slit crack using COMSOL's crack feature. Its orientation is inherited from the crack geometry, and two J-integral features are defined so that the two crack tips can be evaluated separately.

5.3 Boundary Conditions and Loading

The benchmark setup uses a roller constraint on the bottom edge and a zero horizontal displacement condition at the bottom-right corner to remove rigid-body motion. A uniform vertical tensile boundary load of 100 MPa is applied on the top boundary. Because one side is constrained while the opposite side is loaded, the numerical configuration is not perfectly symmetric for inclined cracks; consequently, the left- and right-tip fracture parameters can differ slightly.

5.4 J-Integral Evaluation

Both J-integral features use circular contour integration. The contour radius is set to 0.5 times the crack size, with the lower and upper contour bounds defined as 0.2 and 0.5 times the crack size. A fixed 1000 integration points are used for the contour calculation. The model also evaluates $KI/K0$ and $KII/K0$ for each crack tip.

6. Mesh and Solution Strategy

The model uses a user-controlled triangular mesh. A general size operation is combined with an extra-fine local size specification and a free-triangular mesh operation. The local maximum element size is linked to the crack length through $h_{\max} = a/20$, providing additional refinement in the fracture-critical region.

Mesh setting	Model value
Mesh type	User-controlled
General maximum element size	8.38
Local maximum element size	$a/20$
Predefined local size	Extra fine
Element type	Quadratic serendipity
Mesh operation	Free triangular
$\beta = 67.5^\circ$ elements	796
$\beta = 22.5^\circ$ elements	802
$\beta = 90^\circ$	Stored solution; benchmark model mesh

The study is stationary and includes a parametric sweep over $\beta = 90^\circ$, 67.5° , and 22.5° . The stored solver history shows that the three parameter values were solved successfully in COMSOL 6.3. The reported computation time in the model metadata is approximately 3 s.

7. Results and Discussion

7.1 J-Integral Results

β (deg)	J, left tip (J/m ²)	J, right tip (J/m ²)	Average J (J/m ²)
90.0	5019.879	5021.226	5020.553
67.5	4140.973	4298.066	4219.519
22.5	655.833	676.817	666.325

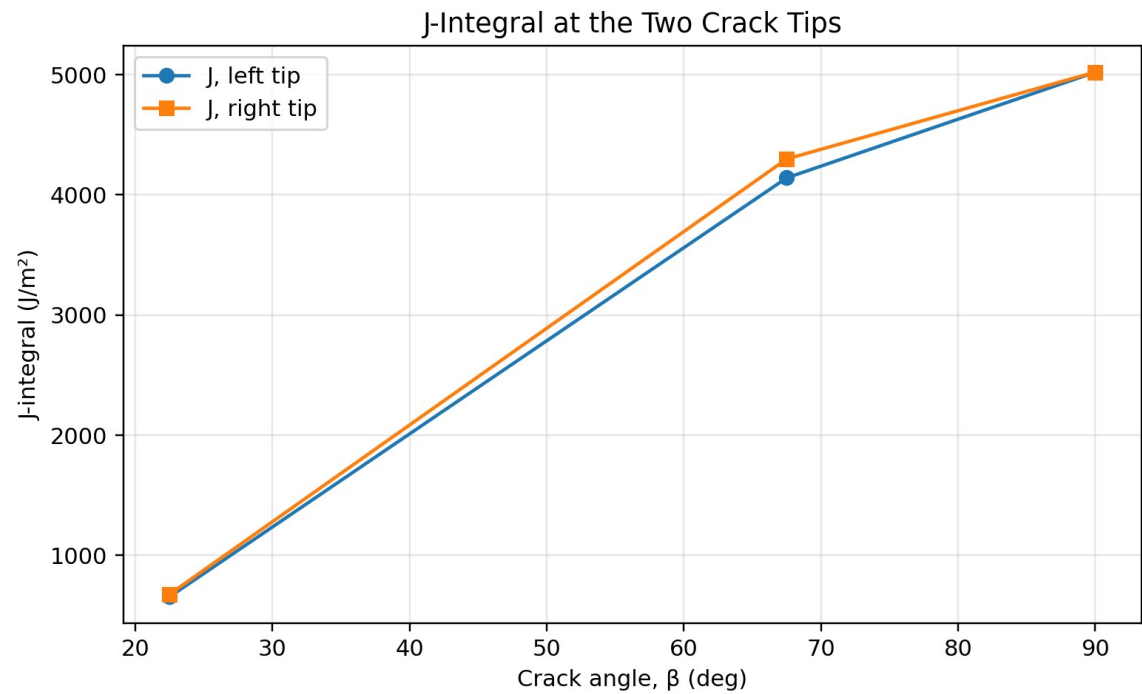


Figure 2. J-integral values obtained at the two crack tips.

The J-integral is highest for $\beta = 90^\circ$, where the crack is horizontal relative to the vertical tensile loading direction. As the crack angle decreases, the average J-integral decreases substantially. The average values are approximately 5020.6 J/m² at 90° , 4219.5 J/m² at 67.5° , and 666.3 J/m² at 22.5° . The two crack tips show close agreement for $\beta = 90^\circ$ and 22.5° , while a larger difference is visible at 67.5° because of the nonsymmetric boundary conditions.

7.2 Mode-I Stress Intensity Factor

β (deg)	KI/K0 left	KI/K0 right	Reference KI/K0	Average error
90.0	1.2057	1.2059	1.200	0.48%
67.5	1.0281	1.0534	1.030	1.04%
22.5	0.1856	0.1886	0.190	1.53%

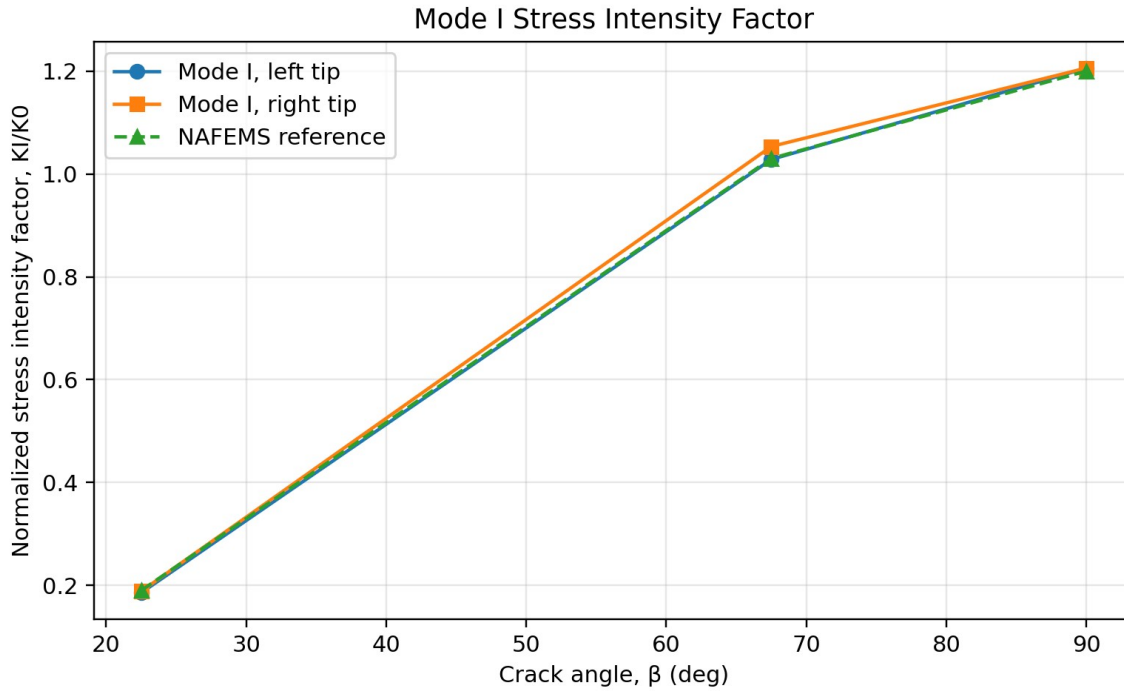


Figure 3. Normalized Mode-I stress intensity factors and NAFEMS reference values.

At $\beta = 90^\circ$, K_I/K_0 is approximately 1.206 at both tips, indicating strong opening-mode behavior. At $\beta = 67.5^\circ$, K_I/K_0 decreases to approximately 1.03–1.05. At $\beta = 22.5^\circ$, it becomes approximately 0.186–0.189. The computed values remain close to the reference values, with average deviations of about 0.48%, 1.04%, and 1.53% for the three angles, respectively.

7.3 Mode-II Stress Intensity Factor

β (deg)	K_{II}/K_0 left	K_{II}/K_0 right	Reference K_{II}/K_0	Average error
90.0	0.0119	-0.0114	0.000	Near zero
67.5	0.3772	0.3678	0.370	0.67%
22.5	0.3943	0.4006	0.405	1.86%

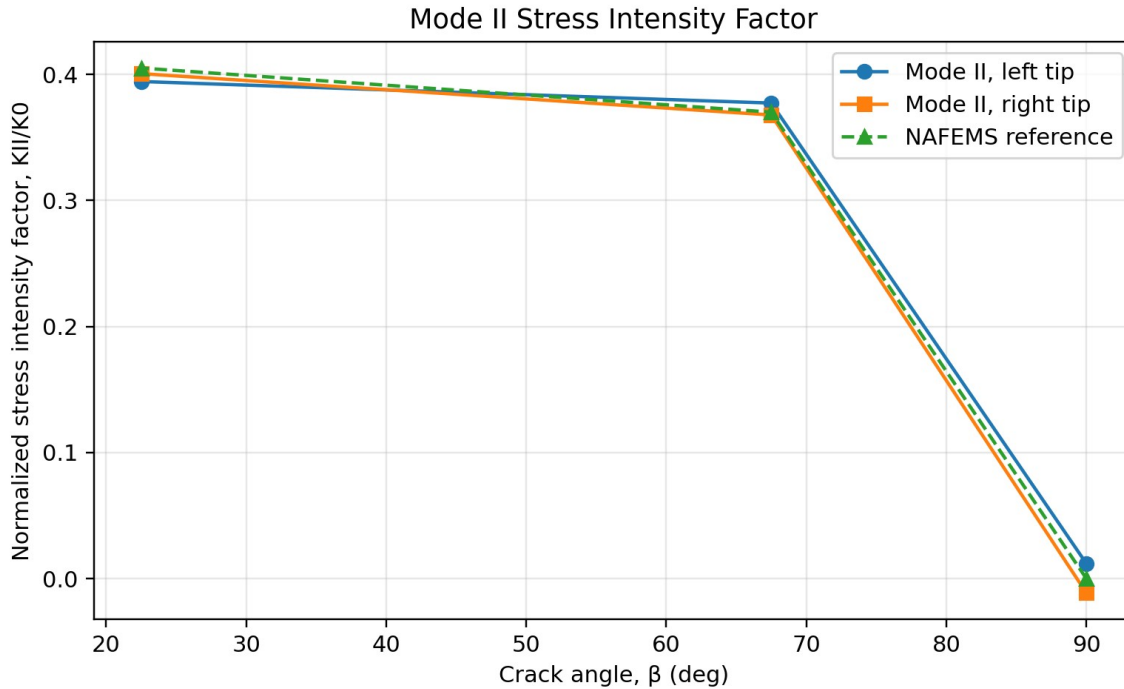


Figure 4. Normalized Mode-II stress intensity factors and NAFEMS reference values.

The Mode-II response is almost zero for $\beta = 90^\circ$, confirming a nearly pure opening condition. At $\beta = 67.5^\circ$, K_{II}/K_0 rises to approximately 0.37. At $\beta = 22.5^\circ$, it increases to approximately 0.40 while K_I/K_0 falls below 0.19. Therefore, the lower-angle configuration becomes Mode-II dominant. The small sign difference between the two 90° -case tip values is numerical/nonsymmetric rather than a physically significant sliding mode.

7.4 Crack-Tip Stress Intensity Factors in Physical Units

β (deg)	KI left (MPa $\sqrt{m}$)	KI right (MPa $\sqrt{m}$)	KII left (MPa $\sqrt{m}$)	KII right (MPa $\sqrt{m}$)
90.0	33.790	33.795	0.334	-0.318
67.5	28.813	29.521	10.571	10.307
22.5	5.201	5.285	11.051	11.226

Using $K_0 = 28.025$ MPa $\sqrt{m}$, the largest Mode-I value occurs for $\beta = 90^\circ$ at approximately 33.79 MPa $\sqrt{m}$. At $\beta = 22.5^\circ$, the Mode-II values of approximately 11.05–11.23 MPa $\sqrt{m}$ exceed the corresponding Mode-I values of approximately 5.20–5.29 MPa $\sqrt{m}$, demonstrating the transition to sliding-dominated fracture.

8. Verification Against the NAFEMS Benchmark

The model is intended as a reproduction of the NAFEMS angle-crack benchmark. The benchmark reports normalized stress intensity factors for the three crack angles. The supplied COMSOL results reproduce those values closely.

β	KI/K0 ref.	KI/K0 avg.	KI error	KII/K0 ref.	KII/K0 avg.	KII error
90.0	1.200	1.2058	0.48%	0.000	0.0003	—
67.5	1.030	1.0407	1.04%	0.370	0.3725	0.67%
22.5	0.190	0.1871	1.53%	0.405	0.3975	1.86%

The agreement is strong across all three angles. For $\beta = 67.5^\circ$ and 22.5° , the average normalized Mode-II errors are approximately 0.67% and 1.86%, respectively. The small discrepancy between left and right crack tips is

consistent with the asymmetric boundary conditions used in the benchmark configuration. Overall, the supplied model provides a successful numerical verification of the expected mixed-mode fracture response.

9. Conclusions

- A two-dimensional finite-element model of a centrally embedded angle crack was successfully documented from the supplied COMSOL Multiphysics file.
- The plate dimensions are 100 mm × 125 mm, with a 50 mm long crack and a nominal tensile load of 100 MPa.
- The J-integral decreases strongly as the crack angle changes from 90° to 22.5°, with average values of approximately 5021, 4220, and 666 J/m².
- For $\beta = 90^\circ$, the fracture response is predominantly Mode I and KII is essentially zero.
- Reducing the crack angle introduces Mode II loading; at $\beta = 22.5^\circ$, the Mode-II stress intensity factor becomes larger than the Mode-I factor.
- The computed normalized stress intensity factors agree closely with the NAFEMS reference values, generally within about 2% when the two crack-tip results are averaged.
- Small differences between the two crack tips arise from the nonsymmetric constraint/loading arrangement and are consistent with the benchmark model.

10. Future Scope

- Perform a systematic crack-angle sweep using smaller angular increments to establish continuous KI/KII trends.
- Conduct a mesh-convergence study focused on crack-tip refinement and J-integral contour independence.
- Investigate the influence of crack length, plate dimensions, and applied stress on the fracture parameters.
- Extend the study to material nonlinearity, elastic-plastic fracture mechanics, or cohesive crack/damage formulations.
- Compare the finite-element results with experimental measurements of crack opening and sliding displacement.

Data provenance note: Numerical parameter values, mesh settings, crack angles, and result-table values in this report were extracted from the supplied angle_crack_plate.mph file. The NAFEMS reference values and benchmark description were checked against COMSOL's official model documentation.